

Root Cause Analysis (RCA) Document

Incident: Users are Unable to Login to InstaSafe Portal

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1. Incident Summary

The InstaSafe Portal experienced a login outage where multiple users were unable to authenticate. The issue prevented access to protected applications and increased support requests.

2. Impact Analysis

- Users could not access the portal.
- Business operations were disrupted.
- Support tickets increased significantly.
- Administrative activities were delayed.

3. Timeline

09:00 - Issue reported by users.
09:10 - Support confirmed authentication failures.
09:20 - Investigation started.
09:35 - SSL certificate found expired.
09:45 - Certificate renewed and service restarted.
10:00 - Login functionality restored.

4. 5 Whys Analysis

Problem: Users cannot log in.

1. Authentication requests failed.
2. Identity Provider could not establish secure communication.
3. HTTPS connections were rejected.
4. The SSL certificate had expired.
5. No automated certificate monitoring or renewal process existed.

5. Root Cause

The authentication service SSL certificate expired, causing secure authentication requests to fail. The absence of certificate monitoring and automated renewal allowed the certificate to expire unnoticed.

6. Resolution

- Renewed the SSL certificate.
- Restarted the authentication service.
- Validated successful logins.
- Monitored system stability.

7. Preventive Actions

- Enable automatic SSL certificate renewal.
- Configure certificate expiry alerts.
- Perform monthly security audits.

- Maintain an incident response checklist.
- Deploy backup authentication infrastructure.

8. Lessons Learned

Critical authentication components should be monitored continuously. Preventive monitoring and documented maintenance procedures reduce downtime.

9. Conclusion

The issue was resolved by renewing the expired SSL certificate and restoring the authentication service. Automated renewal and monitoring will help prevent similar incidents.